

Lecture 6

Nontrivial Clopen Subsets

- $M = (0, 1) \cup (2, 3), A = (0, 1)$
- $M = \mathbb{Q}, A = (e, \pi) \cap \mathbb{Q}$

Definitions of Continuity

- Convergence: Sequence (x_n) in M converges to a limit $p \in M$ means $\forall \varepsilon > 0 \exists N_0 \in \mathbb{N}$ s.t. $\forall n \geq N_0, d(x_n, p) < \varepsilon$
- Sequence Definition: $f : M \rightarrow N$ is continuous at $p \in M$ iff every sequence (x_n) in M that converges to p has a limit $\lim(f(x_n)) = f(p)$
- Epsilon-Delta Definition
- Open Set Condition Definition: $f : M \rightarrow N$ is continuous iff $\forall$ open set $V \subset N$, we have $f^{\text{pre}}(V)$ is open in M .
 - Remark on notation for $f^{-1}(V)$. Often written as $f^{-1} \circ f = \text{id}$. However, f^{-1} isn't always a function, except when f is bijective ($f^{-1} = f^{\text{pre}}$).
 - If f is not injective, $f^{\text{pre}}(\{y\})$ can be a set of points.

Theorem 1: $f : M \rightarrow N$ is continuous w.r.t. $\varepsilon, \delta \Leftrightarrow$ continuous w.r.t. sequence definition.

- Use contrapositive to prove
- What does not continuous w.r.t sequences imply?
- $\exists x_n \rightarrow p \in M$ but $f(x_n) \nrightarrow f(p)$
 - Note here I have traditionally used $\rightarrow$ for implication, but Pugh generally uses $\Rightarrow$. $\rightarrow$ will denote convergence.
- $\exists r > 0$ s.t. $N_{r(f(p))}, \exists N$ s.t. $\forall n > N, f(x_n) \in N_r(f(p))$
- $\exists (x_{n_k})$ subsequence $f(x_{n_k})$ s.t. $\forall k f(x_{n_k}) \notin N_r(f(p))$
- f being continuous means it preserves convergence and limits

Theorem 2: $f : M \rightarrow N$ is continuous iff $\forall$ open $V \subset N, f^{\text{pre}}(V)$ is open in M

- Assume ε, δ -continuous
- $\varepsilon - \delta$ gives us that for all $N_\varepsilon(f(p))$ we have some $M_\delta(p)$ such that $f(M_\delta(p)) \subset N_\varepsilon(f(p))$
- An open set V consists of the union of r -nbh of all points inside
- Each r -nbh has a corresponding nbh in M , which is guaranteed by ε, δ

Homeomorphisms

- $f : M \rightarrow N$ is bijective, continuous, and so is f^{-1}
- Example: $f : [0, 2\pi) \rightarrow \text{circle}, \theta \mapsto (\cos(\theta), \sin(\theta))$
 - Not a homeomorphism, since inverse has some issues as values converge to $(1, 0)$ from the 4th quadrant
 - Values converge to 2π instead of intended 0

Theorem 3: If M, N are compact and $f : M \rightarrow N$ is a continuous bijection, then $f^{-1} : N \rightarrow M$ is a continuous bijection

Theorem 4: Homeomorphisms preserve topologies

- Homeomorphism $f : M \rightarrow N$
- $\mathcal{T}_M \rightarrow \mathcal{T}_N$

- Open set $U \mapsto f(U)$
- Obvious from open set condition
- $M \cong N$ means homeo

Theorem 5: If M is compact and $f : M \rightarrow N$ is continuous, $f(M)$ is a compact subset of N

- Let (y_n) be a sequence in $f(M)$
- $\exists(x_n) \in M$ s.t. $f(x_n) = y_n$
- $\exists x_{n_k} \rightarrow x \in M \Rightarrow f(x_{n_k}) \rightarrow f(x)$ as $k \rightarrow \infty$
- $f(M)$ is a compact subset as any sequence contains a subsequence which converges to a limit in $f(M)$
- Theorem can be extended to homeomorphisms

Theorem 4 Variation: Continuous f instead of Homeo f

- $\mathcal{T}_M \leftarrow \mathcal{T}_N$
- $f^{\text{pre}(V)} \in \mathcal{T}_M \leftarrow V \in \mathcal{T}_N$
- Consequence to contemplate: $N \subset M, \mathcal{T}_N = \{N \cap U : U \in \mathcal{T}_M\}$ $N(r, p) = N \cap M_r(p)$